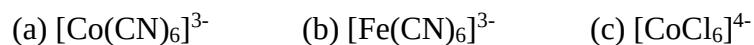


ACML 101 Inorganic Part
Home assignment -1
Crystal Field Theory and its applications to metal complexes

Name: Entry No. Group.....

Please answer all the questions in the provided space and submit them by **December 7, 2024**.

1. Calculate crystal field stabilization energy (CFSE) for the following complexes



2. Draw the crystal field splitting diagram of d orbitals in a linear complex MX_2 , assuming the ligands (X) to be along the z axis.

3. Arrange $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ in decreasing order of metal ion radii with justification.

4. $K_3[Co(CN)_6]$ is colorless while $K_3[Co(C_2O_4)_3]$ is colored. Why?

5. Determine if $NiFe_2O_4$ is a normal or inverse Spinel?

6. Which of the following octahedral complexes are expected to show very weak tetragonal distortion and why?

(a) $[Cr(CN)_6]^{4-}$ (b) $[Mn(H_2O)_6]^{2+}$ (c) $[Cr(H_2O)_6]^{2+}$ and $[Ti(H_2O)_6]^{3+}$